Computer Vision-Based Assessment of Autistic Children: Analyzing Interactions, Emotions, Human Pose, and Life Skills

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- Detecting ASD in children typically relies on methods like clinical observations, standardized questionnaires, and sometimes invasive procedures.
- However, these traditional diagnostic approaches can be time-consuming and subjective, often varying based on the clinician's experience and expertise.
- Thus, this seminar addresses a pioneering approach to assessing autistic children through computer vision and deep learning techniques.

Title	Author	Summary
AI-assisted Initiation to Joint Attention Evaluation for Autism Spectrum Disorder Detection - IEEE 2022	Ko Chany-oung, Kang Soyeon et al	This paper evaluates the performance of joint attention models using publicly available datasets, providing benchmarks for future research in ASD assessment.
Facial Expression Recognition: A Comprehensive Review - IEEE 2021	Vipan Verma, Rajneesh Rani	A review of various methods for facial expression recognition, discussing their applications in different fields, including autism diagnosis.

Title	Author	Summary
Autism Spectrum Disorder Detection Using Machine Learning Approach - IEEE 2021	Nabila Zaman, Jannatul Ferdus et al	This paper discusses the integration of machine learning techniques in the diagnosis of autism, highlighting their effectiveness in improving diagnostic accuracy.
Human Body Pose Estimation and Applications - IEEE 2021	Amrutha K, Prabu P et al	A survey of state-of-the-art methods for human pose estimation, relevant for assessing social interactions in children with ASD.

- Increasing prevalence of ASD.
- Limitations of subjective and time-consuming traditional methods.
- Importance of objective measurements and real-time feedback.

Problem Statement

To create a scalable, efficient, and comprehensive assessment tool that addresses the limitations of current methods and improves outcomes for children with ASD by using video data and advanced machine learning techniques.

Objectives

- Develop automated assessment tools.
- Enhance diagnostic accuracy.
- Facilitate real-time analysis.
- Support comprehensive behavioral evaluation.
- Promote scalable and efficient assessments.
- Contribute to ongoing research.

Methodology Overview

Data Collection

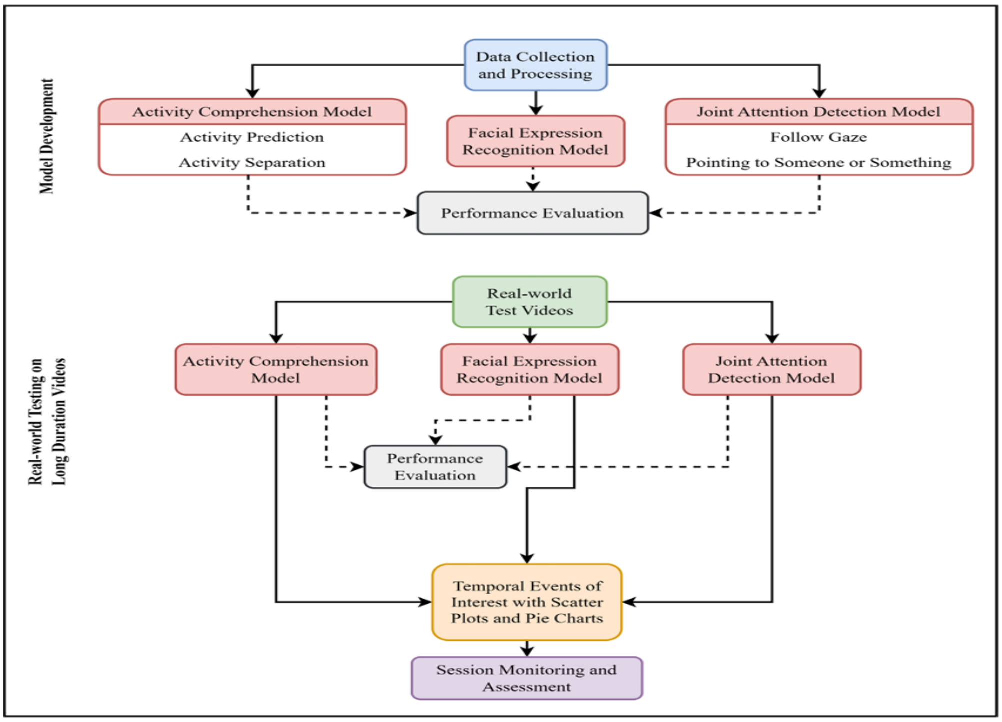
- Description of the dataset (300 videos of ASD children).
- Types of interactions captured (play-based sessions).

Model Development

- Activity Comprehension Model.
- Joint Attention Recognition Framework.
- Facial Expression Recognition Model.

** each model has been tested with 68 unseen videos**

System Architecture



Data Collection

Environment

- Controlled environments - Therapy rooms.
- Naturalistic settings - Home, Playground etc.
- Consent obtained from parents/guardians.
- Ensured anonymity of participants in all data handling and reporting.

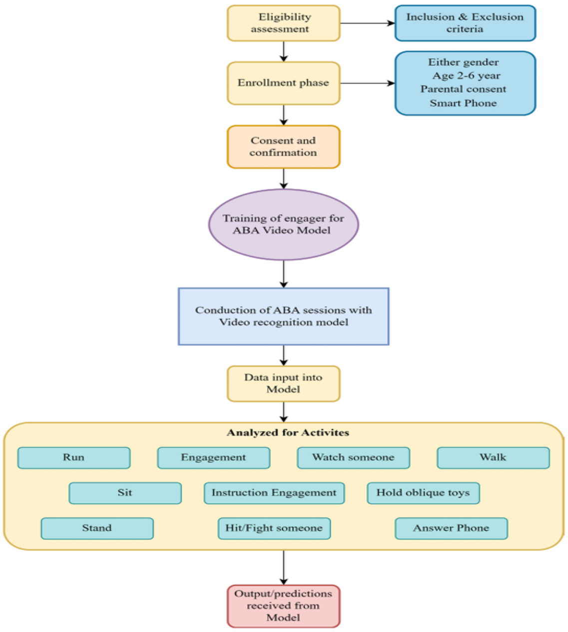
Equipments used

- High-definition cameras - for capturing video data of 30 minutes each.
- Depth sensors.

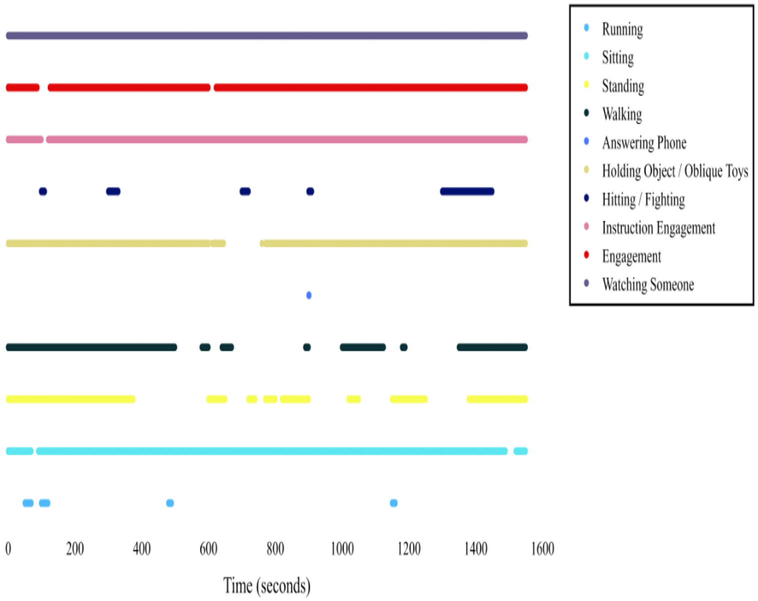
Activity Comprehension Model

- Deep learning algorithms - CNN, RNN.
- Object detection model - YOLO.
- Trained on labeled data to recognize and classify ten distinct activities.
- Model has been tested and identified activities such as running, sitting, engagement with play partner, instruction engagement etc.
- Scattered plots showing activity and non activity periods.
- Overall accuracy of 72.32%.

Model Development



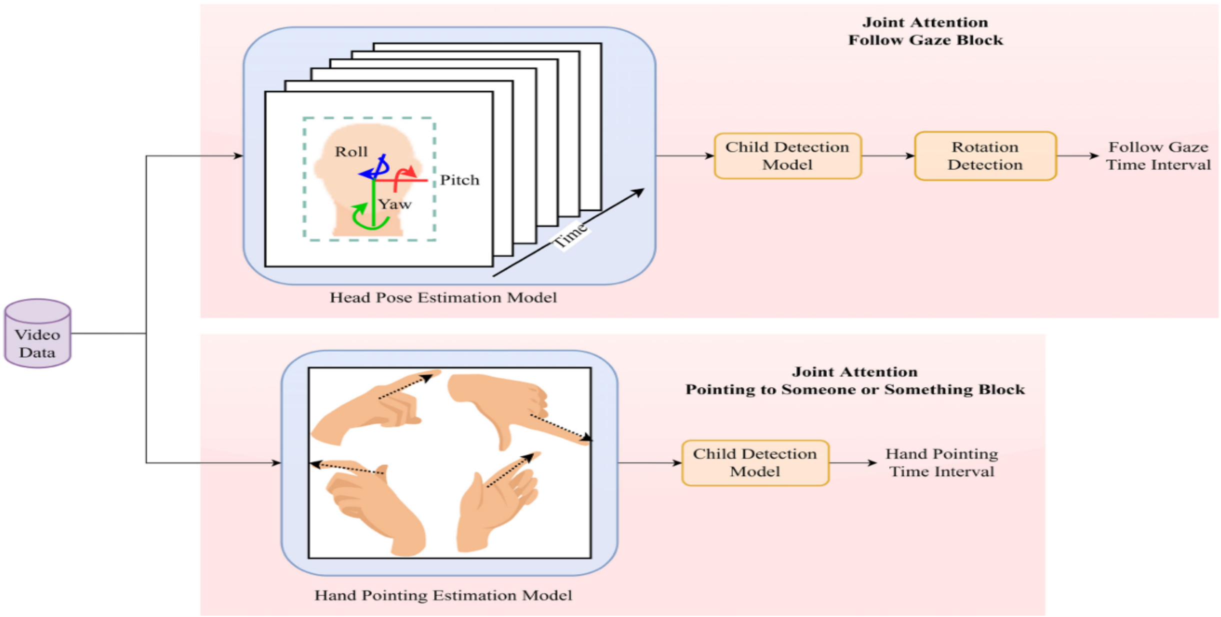
Activity Comprehension Model



Scatter plots of hitting and running

Joint Attention Recognition Framework

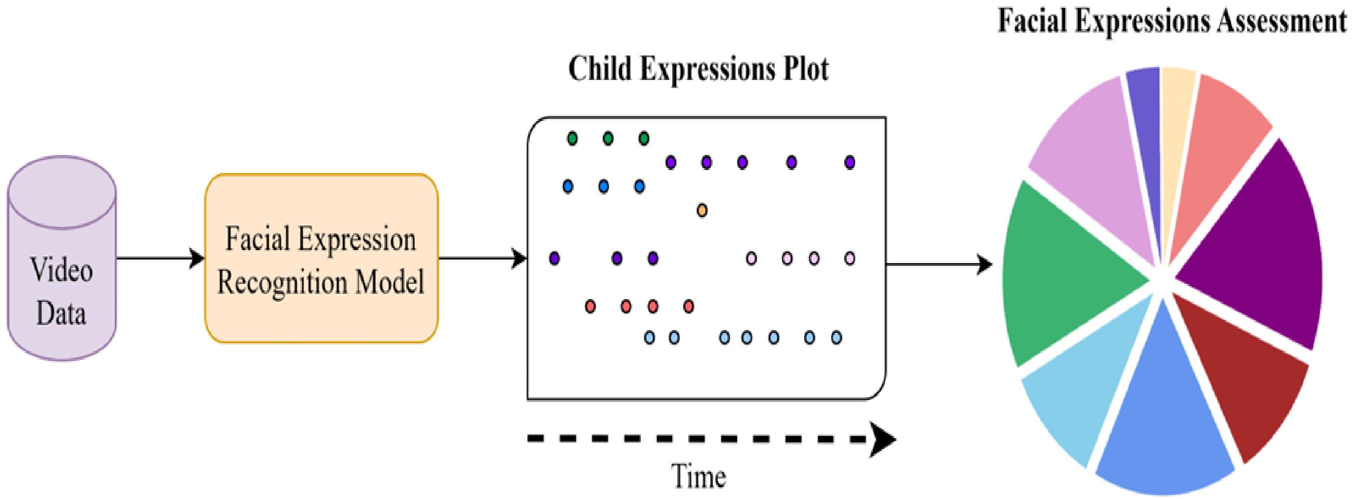
- Deep neural network models designed for joint attention detection.
- Integration of head and hand pose estimation techniques to evaluate responses to social cues.
- Trained on labeled data to recognize two critical types of joint attention:
  1. Follow-Gaze: Child follows the therapist's gaze to an object or location.
  2. Hand Pointing: Child responds to the therapist's hand pointing towards an object.
- Model has been tested and identified instances of follow-gaze and responses to hand pointing.
- Visualised through metrics showing accuracy and frequency of responses.
- Overall accuracy of 97% and 93.4% respectively.



Joint Attention Recognition Framework

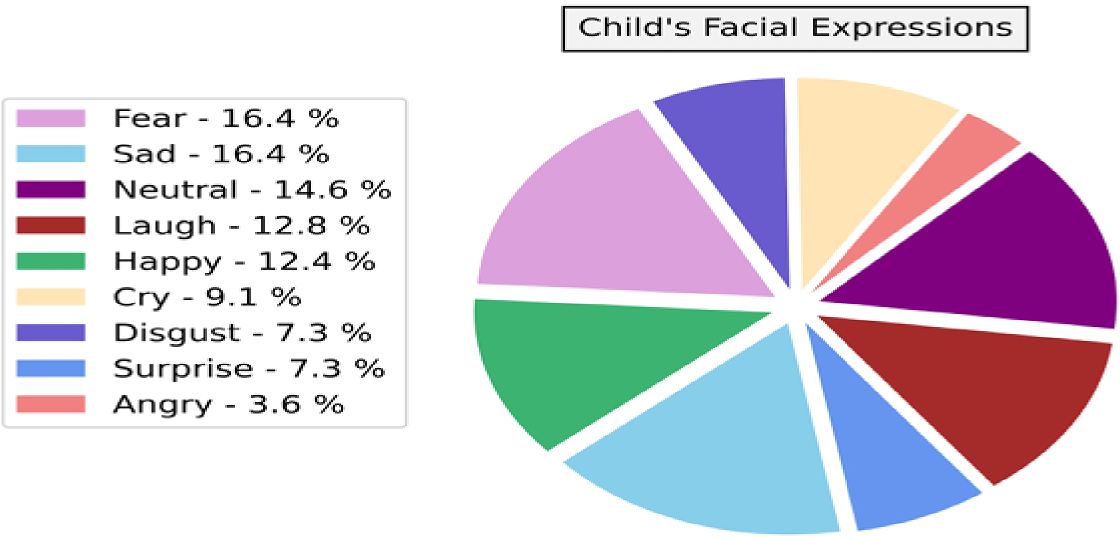
Facial Expression Recognition Model

- Deep learning architectures - CNN.
- Trained on labeled data to recognize facial expressions and classify various emotional states.
- Model has been tested and identified various emotional states like happiness, sadness, anger, surprise etc.
- Visualised through confusion matrices and pie charts, indicating the frequency of detected emotions over time.
- Overall accuracy of 95.1%.



Assessment with FER Model

Model Development



Assessment output

Overall Accuracy

- Activity Comprehension Model-72.32%
- Joint Attention Recognition Framework
  - Follow eye gaze accuracy-97%
  - Hand pointing accuracy-93.4%
- Facial Expression Recognition Model-95.1%

Evaluation of real world videos

- Tested on a diverse set of unseen videos to assess robustness.
- Performance metrics indicate strong generalization capabilities across different settings.

Key Insights

- Scatter plots - clear visual representation of engagement levels over time.
- Discontinuities in the scatter plot - indicate periods of non-engagement.
- Classifies various activities for a nuanced understanding of child behavior during sessions.
- Predictions can inform therapists about specific behaviors

Discussions

- The framework can significantly enhance the assessment process in clinical settings, providing objective data to support therapy decisions.
- Facilitates a more personalized approach to intervention by identifying specific areas of concern.

Challenges

- Difficulty in detecting individuals in crowded or occluded environments.
- Variability in video quality and lighting conditions may affect model performance.

Future Work

- Integration of Multimodal Data - Combine visual and auditory features
- Enhanced model architecture - Multitask Learning Framework
- Expansion of Dataset - Diverse and Larger Datasets
- Longitudinal Studies - Track Progress Over Time

Conclusion

- The research successfully developed and tested computer vision models for assessing children with Autism Spectrum Disorder (ASD).
- Advances the integration of computer vision and deep learning in the assessment and treatment of ASD.
- It highlights the potential for technology to support clinicians in making timely and accurate diagnoses.
- The findings underscore the transformative potential of technology in improving the lives of children with ASD.

References

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Thank You